EP3120 - Statistical Mechanics

Kirit Makwana
Department of Physics, IIT Hyderabad

Ising model

Model for ferro-magnetism

- Materials typically do not have a net magnetization
- If an external magnetic field is applied then a paramagnetic substance develops magnetization parallel to the externally applied field
- A diamagnetic material develops magnetization in opposite direction to the applied magnetic field
- Ferromagnetism is a phenomenon in which some metals (e.g. Iron and Nickel) develop a spontaneous magnetization - i.e., in the absence of an externally applied field
- The hysteresis phenomenon demonstrates this spontaneous magnetization
- However, this happens only under the Curie temperature
- The Ising model tried to explain this from a microscopic point of view
- It is taken as a standard example to study phase transitions

Ising model

- The Ising model considers a lattice of N fixed points
- The spin at each lattice site is s_i, $i = 1, 2, ..., N$ with $s_i = \pm 1$
- A given set of $\{s_i\}$ gives a microstate of this system
- The kinetic energy is neglected. Only the interaction energy between neighboring lattice sites is considered
- The interaction between only nearest neighbors is considered and also between a spin site and an external magnetic field
- The energy of a particular configuration is

$$E\{s_i\} = -\epsilon \sum_{\langle ij \rangle} s_i s_j - B \sum_{i=1}^{N} s_i \tag{1}$$

- ϵ is the energy of interaction. If $\epsilon > 0$ it is ferromagnetism, otherwise it is a model for anti-ferromagnetism

Ising model

- B is the external magnetic field and ϵ is the potential energy of interaction between nearest neighbors
- The symbol $\langle ij \rangle$ represents sum over all the pairs of nearest neighbors, without repetition
- There is no distinction between $\langle ij \rangle$ and $\langle ji \rangle$
- Let γ represent the number of nearest neighbors of a lattice site
- $\gamma = 4$ for a 2D square lattice, $\gamma = 6$ for a simple cubic lattice, $\gamma = 8$ for a body centered cubic lattice
- Therefore, the number of terms in the summation over $\langle ij \rangle$ will be $\gamma N/2$
- The canonical partition function will be

$$Q(B, T) = \sum_{\{s_i\}} e^{-\beta E\{s_i\}} \tag{2}$$

- it will be a function of the temperature and external magnetic field B

Magnetization

- The Helmholtz free energy will be

$$A = -k_B T \ln Q \tag{3}$$

- The average magnetization will be given as

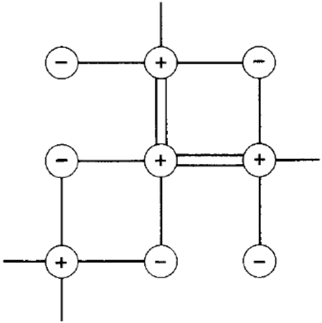
$$M(B, T) = N\langle s_i \rangle = -\frac{\partial}{\partial B}\left(\frac{A}{k_B T}\right) \tag{4}$$

- If we get a non-zero $M(0, T)$ that means we have spontaneous magnetization
- Clearly there will be many different microstates with the same energy, they will be degenerate
- Lets define N_+ and N_- as the number of lattice sites with spins up and down respectively
- Then we have

$$N_- = N - N_+ \tag{5}$$

Nearest neighbor pairs

- There are three different types of nearest neighbor pairs - $(+,+)$, $(-,-)$ and $(+,-)$
- The pair $(-,+)$ pair is the same as $(+,-)$ pair
- Let N_{++}, N_{--}, and N_{+-} denote the number of $(+,+)$, $(-,-)$, and $(+,-)$ pairs respectively
- Consider a lattice site with spin up and draw lines to each of its γ nearest neighbors
- A total of γN_{+} lines will be drawn



Nearest neighbour pairs

- There will be two lines drawn between $(+,+)$ pairs, one line between $(+,-)$ pairs, and no line drawn between $(-,-)$ pairs
- So we can relate

$$\gamma N_+ = 2N_{++} + N_{+-} \tag{6}$$

- Similar consideration can be applied to lattice sites with spin down to obtain

$$\gamma N_- = 2N_{--} + N_{+-} \tag{7}$$
$$N = N_+ + N_- \tag{8}$$

- We can use these three equations to eliminate three variables N_{+-}, N_-, and N_{--} and keep only three other variables N, N_+, and N_{++}

$$N_{+-} = \gamma N_+ - 2N_{++} \tag{9}$$
$$N_- = N - N_+ \tag{10}$$
$$N_{--} = \frac{\gamma}{2}N + N_{++} - \gamma N_+ \tag{11}$$

Energy

- From the number of nearest neighbor pairs we get

$$\sum_{\langle ij \rangle} s_i s_j = N_{++} + N_{--} - N_{+-} \tag{12}$$

$$= 4N_{++} - 2\gamma N_{+} + \frac{\gamma}{2} N \tag{13}$$

$$\sum_i s_i = 2N_{+} - N \tag{14}$$

- Substituting these expressions in the energy we get

$$E\{s_i\} = -\epsilon(4N_{++} - 2\gamma N_{+} + \frac{\gamma}{2} N) - B(2N_{+} - N) \tag{15}$$

$$= -4\epsilon N_{++} + 2(\epsilon\gamma - B)N_{+} - (\frac{\epsilon\gamma}{2} - B)N \tag{16}$$

- This shows that the energy for N lattice points depends only on two details of the configuration - N_{++} and N_{+}

Partition function

- The partition function then becomes

$$Q = \sum_{s_1}\sum_{s_2}\dots\sum_{s_N} e^{-\beta E\{s_i\}} \tag{17}$$

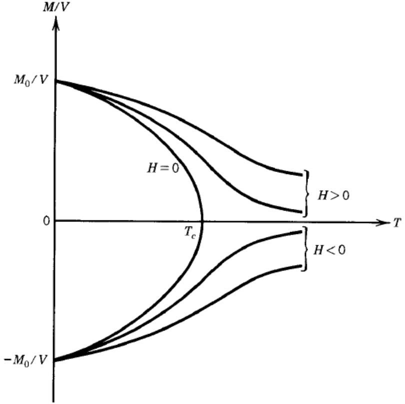
$$= e^{\beta N(\epsilon\gamma/2-B)} \sum_{s_1}\dots\sum_{s_N} e^{-2\beta N_+(\epsilon\gamma-B)} e^{4\beta\epsilon N_{++}} \tag{18}$$

$$= e^{\beta N(\epsilon\gamma/2-B)} \left[\sum_{N_+=0}^{N} e^{-2\beta(\epsilon\gamma-B)N_+} \left(\sum_{N_{++}} g(N_+, N_{++}) e^{4\beta\epsilon N_{++}} \right) \right] \tag{19}$$

- Here the function $g(N_+, N_{++})$ represents the number of different possible configurations of the lattice with a given N_+ and N_{++}
- The second summation will run over all values of N_{++} that are consistent with N sites with N_+ spins up
- Various ways have been devised to evaluate and approximate this function

Spontaneous magnetization

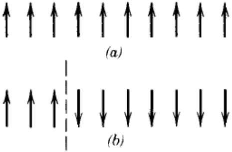
- The magnetization per unit volume of a ferromagnetic substance



- T_c is the Curie temperature
- There is a spontaneous magnetization below T_c

Magnetization in 1D

- Consider a 1D lattice in which the spins are all aligned



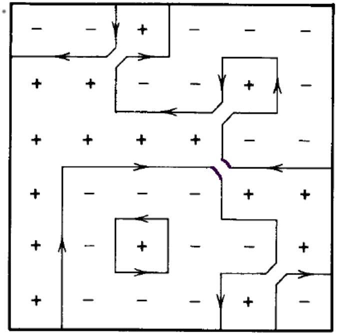
- In this case the energy is minimum, but entropy is also minimum
- We can create a partition and flip all spins to the right of this partition
- This changes the free energy by

$$\Delta A = \Delta U - T\Delta S = 2\epsilon - Tk_B \ln(N-1) \tag{20}$$

- For finite temperature and large N, this means that every partition will increase the free energy
- This implies that there can be no spontaneous magnetization at finite temperature in 1D

Magnetization in 2D

- Consider a 2D square lattice of spins
- Let us draw domain walls between regions of up and down spins
- These walls can start and end on the boundary or they can be closed loops inside
- They have a direction such that the down spins are always to the right and up spins are to the left
- If there is an ambiguity, the wall will bend to the right



Magnetization in 2D

- The length of a domain wall is given by the number of lattice spacings it traverses
- Following these rules no two domain walls will cross
- Two domain walls of same shape but different location are distinct
- Two domain walls of same shape and location, but different oriantation are also counted as distinct
- The different microstates map to the different configurations of domain walls
- We impose a boundary condition that all sites on the boundary are spin up - in the limit of large N the contribution of this boundary condition to the average magnetization will tend to zero
- If we are able to find a finite non-zero average magnetization in the limit of large N and $B \to 0$ then we can claim that spontaneous magnetization exists
- As a result of this boundary condition, only closed domain walls will be present

Domain walls

- Now we will try to obtain an upper bound on the fraction of N_- lattice sites
- If this upper bound is less than $1/2$ then that means we have spontaneous magnetization
- Lets classify the domain walls by their length b
- Lets label the different domain walls with a specific length b with label i
- Any domain wall is characterized by label (b, i)
- A closed domain wall of length b can have a maximum area of $b^2/16$
- Let $m(b)$ denote the number of distinct domain walls with length b
- Exercise: Let us count the number of ways we can construct distinct domain walls of length b (open, closed, or crossing) on a lattice of N sites

Domain walls

- We can build domain walls by placing N sticks one after another to form a continous line
- The first stick can go in $2N$ locations
- Each subsequent stick can go in 3 directions (ignoring the constraints placed by the boundary), and there are $b-1$ number of such sticks
- The number of ways is

$$2(N)3^{b-1} \tag{21}$$

- Since we have considered all possible domain walls, we can say that the number of closed domain walls with length b has to be

$$m(b) \leq 3^{b-1}N \tag{22}$$

- Since the boundary has all $+$ spins, the $-$ spins have to be internal and they have to be enclosed by at least one domain wall

Domain walls

- In order to get an upper bound on the number of $-$ spins in a given configuration, let us define a function

$$X(b,i) = \begin{cases} 1 & \text{if domain wall (b,i) occurs in this configuration} \\ 0 & \text{otherwise} \end{cases} \tag{23}$$

- Then we can get an upper bound on N_-

$$N_- \leq \sum_b \sum_{i=1}^{m(b)} X(b,i)(b^2/16) \tag{24}$$

- The average N_- spins over different microstates will then have an upper bound

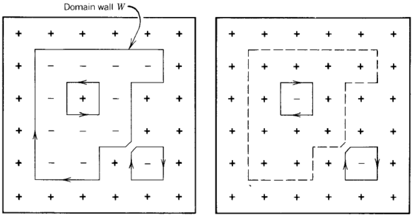
$$\langle N_- \rangle \leq \sum_b (b^2/16) \sum_{i=1}^{m(b)} \langle X(b,i) \rangle \tag{25}$$

Upper bound on N_-

- The average $\langle X(b,i) \rangle$ will be given as

$$\langle X(b,i) \rangle = \frac{\sum_s X(b,i) e^{-\beta E_s}}{\sum_s e^{-\beta E_s}} \tag{26}$$

- Consider a microstate s in which a particular domain wall (b,i) is present
- Then consider another microstate $\tilde{s}$ in which all the spins inside this domain wall are flipped



Upper bound on N_-

- The energy of configuration $\tilde{s}$ will be related to the configuration s as

$$E_s = E_{\tilde{s}} + 2b\epsilon \tag{27}$$

- In the calculation of $\langle X(b,i) \rangle$ let us split the summation into microstates s' which contain the domain wall (b,i) and microstates s'' which do not, so we have

$$\langle X(b,i) \rangle = \frac{\sum_{s'} e^{-\beta E_{s'}}}{\sum_{s'} e^{-\beta E_{s'}} + \sum_{s''} e^{-\beta E_{s''}}} \tag{28}$$

$$\leq \frac{\sum_{s'} e^{-\beta E_{s'}}}{\sum_{s''} e^{-\beta E_{s''}}} \tag{29}$$

- The configuration $\tilde{s'}$ will not have the domain wall (b,i) and so it will belong to the summation in the denominator
- If we restrict the sum in the denominator only to those $\tilde{s'}$ configurations, then the denominator will further decrease and we will get a stronger upper bound

Upper bound on N_-

- Restricting the denominator sum to only those microstates $\tilde{s'}$

$$\langle X(b,i)\rangle \leq \frac{\sum_{s'} e^{-\beta E_{s'}}}{\sum_{\tilde{s'}} e^{-\beta E_{\tilde{s'}}}} \tag{30}$$

$$\leq \frac{\sum_{s'} e^{-\beta E_{s'}}}{\sum_{\tilde{s'}} e^{-\beta E_{s'}} e^{\beta 2b\epsilon}} \tag{31}$$

$$\leq e^{-2b\epsilon\beta} \tag{32}$$

- Substituting in $\langle N_-\rangle$ we have

$$\langle N_-\rangle \leq \sum_b \frac{b^2}{16} \sum_{i=1}^{m(b)} e^{-2b\beta\epsilon} \tag{33}$$

$$\leq \sum_b \frac{b^2}{16} 3^{b-1} N e^{-2b\beta\epsilon} \tag{34}$$

Fraction of magnetization

- The fraction of magnetization becomes

$$\frac{\langle N_- \rangle}{N} \leq \sum_{b=4,6,8,..} \frac{b^2}{48} 3^b e^{-2b\beta\epsilon} \tag{35}$$

$$\leq \frac{1}{48} \sum_{b=4,6,8,..} b^2 (3e^{-\beta\epsilon})^b \tag{36}$$

- The summation goes over $4, 6, 8, ..$ because we are only considering closed loops
- Redefine $b \to 2b$ to get

$$\frac{\langle N_- \rangle}{N} \leq \frac{1}{48} \sum_{b=2,3,4,..} 4b^2 (9e^{-2\beta\epsilon})^b \tag{37}$$

- Define $x \equiv 9e^{-2\beta\epsilon}$

Fraction of magnetization

- We need to evaluate

$$S = \sum_{b=2,3,..} b^2 x^b \tag{38}$$

$$= \sum_{b=1,2,..} (b+1)^2 x^{(b+1)} \tag{39}$$

$$= x\{ \sum_{b=1,..} b^2 x^b + 2 \sum_{b=1,..} b x^b + \sum_{b=1,..} x^b \} \tag{40}$$

$$= x\{ x + S + 2 \sum_{b=1,..} b x^b + \frac{x}{1-x} \} \tag{41}$$

- Now the second term can be evaluated as follows

$$R = \sum_{b=1,..} b x^b = \sum_{b=0,..} (b+1) x^{b+1} = x\{ \sum_{b=1,..} b x^b + \sum_{b=0,..} x^b \} \tag{42}$$

$$= x[R + \frac{1}{1-x}] \tag{43}$$

Fraction of magnetization

- This gives

$$R = \frac{x}{(1-x)^2} \tag{44}$$

- Substituting in the expression for S gives

$$S(1-x) = x^2 + \frac{2x^2}{(1-x)^2} + \frac{x^2}{(1-x)} \tag{45}$$

$$\Rightarrow S = \frac{4x^2}{(1-x)^3}(1 - \frac{3}{4}x + \frac{x^2}{4}) \tag{46}$$

- Substituting this gives

$$\frac{\langle N_- \rangle}{N} \leq \frac{x^2}{3(1-x)^3}(1 - \frac{3}{4}x + \frac{x^2}{4}) \tag{47}$$

Spontaneous magnetization

- If $x = 1/2$ then the fraction on the LHS will be less than $11/24$
- This means that we can always find a finite temperature where the lattice will have a spontaneous magnetization
- That will be the Curie temperature
- This can be proved for a 3D lattice as well
- Lets define the long range order parameter as L

$$\frac{N_+}{N} \equiv \frac{1}{2}(L+1) \tag{48}$$

- This represents the fraction of N_+ sites in the overall lattice
- The short range order parameter is defined as σ

$$\frac{N_{++}}{\gamma N/2} \equiv \frac{1}{2}(\sigma+1) \tag{49}$$

- This represents the fraction of $(+,+)$ links overall